

# K. Arun Kumar

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## Address

2078a, Sector 66,  
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India  
Phone No: +91 9986164687  
email: karun1993@gmail.com

## Positions held

- 2016–20 **Research Assistant(PhD)**, *Universität Osnabrück*, Osnabrück  
part of the graduate school "DFG-GK1916 Combinatorial Structures in Geometry"  
2021–24 **Postdoctoral fellow**, *IISER Tirupati*, Tirupati  
2024–24 **Visiting Scientist**, *ISI Bangalore*, Bangalore  
2024–current **Postdoctoral fellow**, *IISER Mohali*, Mohali

## Education Summary

- 2008–09 **SSE**, *Kendriya Vidyalaya NAL*, Per: 90%  
2010–11 **SSCE**, *Kendriya Vidyalaya NAL*, Per: 93.2%  
2011–16 **B.S-M.S dual degree**, *IISER*, Pune, CGPA:9.4  
2016–20 **PhD in mathematics**, *Universität Osnabrück*, Osnabrück, Grade:Very Good

## Master Thesis

- title Topics in Algebraic K-theory  
supervisor Dr. Amit Hogadi  
description Additivity theorem for Quillen's Q-construction and Waldhausen's S-construction of higher K-groups  
grade Excellent

## Doctoral Thesis

- title On the Motivic spectrum BO and Hermitian K-theory  
supervisor Prof. Oliver Röndigs  
description We extend Panin and Walter's motivic spectrum BO to arbitrary schemes and study its properties  
defense date November 2020  
grade very good

## ■ Courses Taught

08/23-11/23 **Instructor**, *Elementary Number Theory*, IISER Tirupati

## ■ Research Interests

- Algebraic and Hermitian K-theory
- Motivic Homotopy theory
- Equivariant Homotopy theory

## ■ Preprints and publications

- Construction of the motivic cellular spectrum  $KO^{geo}$  over  $Spec(Z)$ , *New York Journal of Mathematics Volume 30 (2024)*, 323-350.
- Endomorphisms of equivariant algebraic K-theory (with Girja Tripathi), *Journal of Homotopy and Related Structures Volume 20 (2025)*, Issue 3.
- A cellular absolute motivic ring spectrum representing Hermitian K-theory (with Oliver Röndigs) (*submitted for publication*)
- Equivariant algebraic cobordism (with Girja Tripathi) (*in preparation*)

## ■ Awards and Recognitions

- Academic excellence Prize: Best academic performance in the 6th semester, B.S-M.S, IISER Pune
- Academic excellence Prize: Best academic performance in the 7th semester, B.S-M.S, IISER Pune
- Rank 8 in CSIR-UGC Test for Junior Research Fellowship held on 21/06/2015

## ■ Recent Conferences and Workshops attended

- Workshop Motivic Algebraic Topology, TIFR, Mumbai, January 6-10th 2025
- Geometric Aspect of Algebraic Varieties (Gaav 2025), IISER Mohali, March 17-19th 2025
- Lean for the Curious Mathematician, ICTS, Bangalore, April 24-26th 2025
- Young homotopy-theorist meeting, ICMS, Edinburgh, October 6-10th 2025

## ■ References

- Prof. Oliver Röndigs, *Universität Osnabrück*  
Email: oroendig(AT)uni-osnabrueck.de
- Dr. Girja Tripathi, *IISER Tirupati*  
Email: girja(AT)labs.iisertirupati.ac.in
- Dr. Chetan Balwe, *IISER Mohali*  
Email: cbalwe(AT)iisermohali.ac.in